

Program : Diploma in Renewable Energy	
Course Code : 5293C	Course Title: Digital Electronics and Microprocessors
Semester: 5	Credits: 4
Course Category : Program Elective Course	
Periods per week: 4 (L:3 T:1 P:0)	Periods per semester: 60

Course Objectives:

- To understand the number systems and logic gates.
- To impart the basic knowledge of combinational logic circuits
- To identify the concepts of sequential logic circuits
- To understand the fundamentals of 8085 microprocessor

Course Prerequisites:

Topic	Course Code	Course name	Semester
Analog Electronics		Analog and Opto Electronics	3

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Identify various number systems and logic families	15	Applying
CO2	Summarize the concepts of combinational logic circuits and flip flops	14	Understanding
CO3	Identify the concepts of Sequential logic and Data conversion circuits	15	Applying
CO4	Identify various instructions and applications of 8085 microprocessor	14	Understanding
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Identify various number systems and logic families		
M1.01	Identify various number systems and their conversions	4	Applying
M1.02	Perform various binary arithmetic operations	4	Applying
M1.03	Solve various boolean expressions using Karnaugh Map	4	Applying
M1.04	Classify various logic gates and logic families	3	Understanding
Contents: Number systems and Conversions -Binary , Octal, Decimal Number system , Hexadecimal numbers-Decimal to binary conversion and vice versa - Decimal to Hexadecimal conversion and vice versa -Hexadecimal to binary conversion and vice versa -ASCII code, BCD numbers- simple problems Binary Arithmetic - Binary addition,subtraction,multiplication-Division - Signed binary number representation-One's complement -Two's complement -Subtraction using one's and two's complements- simple problems			

Boolean algebra and Karnaugh Map - Basic Boolean laws and theorems- Demorgan's Theorem(statement and explanation only)-simplification using Demorgan's theorem -simple problems

Karnaugh Map (K-Map) -Basic concepts - Sum Of Product(SOP) method-Product Of Sum(POS) method- minterm and maxterms- standard SOP expressions- - simplification of SOP expressions using K-map (up to 4 variables)-simple problems

Logic Gates and Logic Families- Logic gates -symbols-truth tables-working with diagrams (basic gates,universal gates and special gates)- universal properties of NAND and NOR gates

Logic families- Characteristics of logic families (noise margin, noise immunity, propagation delay, current sourcing and current sinking, fan in, fan out, power dissipation, speed power product) TTL and CMOS logic families-Comparison of CMOS and TTL performance.

CO2	Summarize the concepts of Combinational logic circuits and Flip flops		
M2.01	Illustrate various combinational logic circuits	4	Understanding
M2.02	Outline the working of multiplexer and demultiplexer	3	Understanding
M2.03	Explain encoder and decoder circuits	3	Understanding
M2.04	Summarize the working of various flip flops	4	Understanding
	Series Test – I	1	

Contents :

Combinational circuits- Arithmetic circuits- Symbol- logic diagram-truth table- basic working- Half adder-Full adder-Half subtractor-Full subtractor

Multiplexers and Demultiplexers-Multiplexers- Symbol- logic diagram-truth table-basic working -4 x 1 MUX, 8x1 MUX-Applications

Demultiplexers-Symbol- logic diagram-truth table-basic working -1 x 4 DEMUX, 1 x 8 DEMUX- Applications

Decoders and Encoders- Symbol- logic diagram-truth table-basic working -3 to 8 Decoder-BCD to decimal Decoder-Decimal to BCD encoder

Flip Flops - Symbol- logic diagram-truth table-basic working-SR-clocked SR- D- JK, -T -JK Master Slave flip flop-Applications of flip flops			
CO3	Identify the concepts of Sequential logic and Data conversion circuits		
M3.01	Summarize various types of shift registers	3	Understanding
M3.02	Identify various counters using flip flops	5	Applying
M3.03	Explain different types of digital to analog converters	3	Understanding
M3.04	Outline various analog to digital converters	3	Understanding
Contents : Shift registers- Types - logic diagram-truth table-basic working-SISO-SIPO-PISO-PIPO - Application of shift registers Counters – classification (asynchronous and synchronous counters) -Modulus of a counter -logic diagram-truth table-basic working-Decade counter- Mod-N asynchronous counter - Mod-N synchronous counter (up to 4 bit)- concept of UP counter,-DOWN counter-applications of counters. Digital to analog converters – Basic DAC techniques- logic diagram-working -Weighted Resistor DAC -R - 2R ladder DAC-applications Analog to Digital Converters - logic diagram-working - Flash type ADC-Successive Approximation converter-Dual slope ADC-applications			
CO4	Identify various instructions and applications of 8085 microprocessor		
M4.01	Outline the fundamentals of Microprocessor	2	Understanding
M4.02	Illustrate the architecture of 8085 microprocessor	5	Understanding
M4.03	Summarize various operations of 8085 microprocessor	2	Understanding
M4.04	Develop simple assembly language programs of 8085 microprocessor	6	Applying
	Series Test – II	1	
Contents : Fundamentals of Microprocessor - Introduction to Microprocessors Evolution of Microprocessors, general block diagram of microprocessor based system- functions of microprocessor.			

8085 Microprocessor -Features - internal architecture- registers -bus structure - Pin diagram-programming model- flag register

8085 instruction classification - Data transfer operations - arithmetic operations -Logical operations -branching operations -machine control operations- Opcode format - data format- instruction format

8085 instruction set - addressing modes - assembly language programming -instruction set - simple programs-selection criteria of microprocessor-applications of microprocessor

Text / Reference

T/R	Book Title/Author
T1	A Anand Kumar, Fundamental of Digital Electronics ,Prentice Hall
T2	S. Salivahanan , Digital Circuits and Design, Oxford University Press
T3	Floyd T.L, Digital Fundamentals, 10/e, Pearson Education, 2011.
T4	D Roy Choudhury & Shail Bala Jain ,Linear Integrated Circuits, New Age International Publications
T5	Ramesh S Gaonkar. Microprocessor architecture programming and applications with the 8085: Prentice Hall
R1	Albert Paul Malvino & Donald P. Leach. Digital principles and applications: McGraw-Hill
R2	Taub & Schilling: Digital Integrated Electronics, McGraw Hill, 1997.

Online Resources

Sl.No	Website Link
1	https://www.electricveda.com/digital-electronics/logic-families-in-digital-electronics-ttl-cmos-and-ecl
2	https://www.elprocus.com/half-adder-and-full-adder/
3	https://www.electronicshub.org/multiplexer-and-demultiplexer/
4	https://www.geeksforgeeks.org/encoders-and-decoders-in-digital-logic/
5	https://www.geeksforgeeks.org/shift-registers-in-digital-logic/
6	https://electronicscoach.com/asynchronous-counter.html
7	https://circuitdigest.com/tutorial/synchronous-counter

8	https://www.elprocus.com/analog-to-digital-converter/
9	https://www.elprocus.com/digital-to-analog-converter-dac-applications/
10	https://www.geeksforgeeks.org/architecture-of-8085-microprocessor/
11	https://www.tutorialspoint.com/microprocessor/microprocessor_8085_instruction_sets.htm
12	https://www.geeksforgeeks.org/addressing-modes-8085-microprocessor/